When a product state stays a product state

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Abstract

Let H be a Hamiltonian on $\mathbb{C}^m \otimes \mathbb{C}^n$ and let $a \otimes b$ be a product state whose evolution $e^{-itH}(a \otimes b)$ is a product state at every time t. We show that the same trajectory is generated by a time-independent local Hamiltonian: there are Hermitian h_A and h_B with $e^{-itH}(a \otimes b) = e^{-ith_A}a \otimes e^{-ith_B}b$ for all real t. No assumption is made on the spectrum of H; degenerate eigenvalues and resonant frequencies are allowed. By induction the same holds for any number of parties. Local generators that depend on time were known to exist; the content here is that a constant one always does. The hypothesis cannot be weakened to constant entanglement: an isotropic exchange coupling in a field keeps the Schmidt coefficients of a suitable two-qubit state fixed for all time, yet no constant local Hamiltonian generates its trajectory; for two qubits that stay maximally entangled, a constant local generator does exist. The proof of the main theorem turns the trajectory into a matrix of polynomials on a torus, factors it in a unique factorisation domain, and reads off orthogonality from a grading argument. The algebraic core and the passage from exponential sums to polynomials are formally verified in Lean.

1 The question

Two qubits interact through $H = Z \otimes Z$, where $Z = \mathrm{diag}(1, -1)$. Start them in $|0\rangle \otimes |+\rangle$, with $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Since $Z|0\rangle = |0\rangle$,

$$e^{-itZ\otimes Z}\big(|0\rangle \otimes |+\rangle\big) = |0\rangle \otimes e^{-itZ}|+\rangle,$$

so the state never becomes entangled, even though H has no local part at all. The motion is exactly what the local Hamiltonian $h_A = 0$, $h_B = Z$ would produce.

A second example has more symmetry. Let $H = J(X\otimes X + Y\otimes Y + Z\otimes Z) + B(Z\otimes 1 + 1\otimes Z)$, the isotropic exchange coupling in a uniform field. The exchange term equals 1 on the symmetric subspace, so for every single-qubit state ϕ,

$$e^{-itH}(\phi \otimes \phi) = e^{-it(BZ+J)}\phi \otimes e^{-itBZ}\phi,$$

again a product at all times, generated by $h_A = BZ + J$ and $h_B = BZ$.

In both examples the interaction is invisible to one particular state, and the trajectory is the trajectory of a local, time-independent Hamiltonian. The question, asked on Physics Stack Exchange [8], is whether this is always so.

Theorem 1. *Let H be a Hermitian operator on $\mathbb{C}^m \otimes \mathbb{C}^n$ and let $a \in \mathbb{C}^m$, $b \in \mathbb{C}^n$ be nonzero. If $e^{-itH}(a \otimes b)$ is a product vector for every $t \in \mathbb{R}$, then there are Hermitian operators h_A on $\mathbb{C}^m$ and h_B on $\mathbb{C}^n$ such that*

$$e^{-itH}(a \otimes b) = e^{-ith_A}a \otimes e^{-ith_B}b \qquad \text{for all } t \in \mathbb{R}.$$

The hypothesis is about one trajectory only; H may entangle every other product state. Nothing is assumed about the spectrum of H. This matters, because the most direct argument breaks down at resonances. If H has eigenvalues $0, 1, 2$ then the frequency 2 arises both as $0+2$ and as $1+1$, and a cancellation between the two could in principle hold a state on the set of product vectors without any local mechanism. The proof below never compares sums of frequencies, and resonances play no role.

Without the words "time-independent" the statement is easy and known. Durt [2] shows that a product state stays a product state exactly when H never couples it to a product state biorthogonal to it, and that it then follows the local but time-dependent mean-field Hamiltonians $H_A^{\mathrm{eff}}(t)$ and $H_B^{\mathrm{eff}}(t)$ obtained by partial traces against the current state of the other party; see also [3]. Sperling and Walmsley [9] derive the same kind of equations of motion for separable trajectories. In these constructions the generator of each factor depends on the state of the other factor and therefore on t. Theorem 1 says that a constant generator always exists. The case where every product state stays a product state is classical: then H itself is a sum of local terms [2].

Two consequences are immediate. The first answers the question as it was asked, for any number of parties; the reduction to two parties was pointed out in the comments of [8] by Norbert Schuch.

Corollary 2. *Let H be Hermitian on $\mathbb{C}^{d_1} \otimes \cdots \otimes \mathbb{C}^{d_N}$ and let $\psi_0 = a_1 \otimes \cdots \otimes a_N$. If $e^{-itH}\psi_0$ is a product vector across all N factors for every t, then there are Hermitian $h_1, \ldots, h_N$, each acting on one factor, with $e^{-itH}\psi_0 = e^{-ith_1}a_1 \otimes \cdots \otimes e^{-ith_N}a_N$ for all t.*

Proof. Induction on N. Apply Theorem 1 to the cut between the first factor and the rest: $e^{-itH}\psi_0 = e^{-ith_1}a_1 \otimes e^{-itH'}(a_2 \otimes \cdots \otimes a_N)$ with H' Hermitian on the last $N-1$ factors. A product vector determines its factors up to scalars, so $e^{-itH'}(a_2 \otimes \cdots \otimes a_N)$ is a product across the last $N-1$ factors for every t, and the induction hypothesis applies to H'. □

Corollary 3. *In the setting of Theorem 1, let K be the smallest H-invariant subspace containing $a \otimes b$. Then H and $G = h_A \otimes 1 + 1 \otimes h_B$ agree on K.*

Proof. The trajectories $e^{-itH}(a \otimes b)$ and $e^{-itG}(a \otimes b)$ coincide, so all their derivatives at $t = 0$ coincide: $H^r(a \otimes b) = G^r(a \otimes b)$ for every $r \geq 0$. These vectors span K, and H maps each of them to $H^{r+1}(a \otimes b) = G^{r+1}(a \otimes b)$, which is where G maps it too. □

So the two examples above are typical: a product trajectory always lives in a subspace on which the Hamiltonian is secretly local.

The route. A product state is a rank-one matrix, so the hypothesis says that a matrix-valued exponential sum has rank one at every time. If all the frequencies were integers, this sum would be a matrix of polynomials in $z = e^{-it}$ evaluated on the unit circle, rank one on the circle and therefore rank one as a polynomial matrix. In general we choose a basis of the frequencies and use one variable per basis element, so the sum becomes a polynomial matrix evaluated along a dense line on a torus (Section 2). A rank-one polynomial matrix factors as a column times a row of polynomials (Section 3), and each factor is a candidate trajectory. What is missing is that each factor moves like a Hamiltonian trajectory, which amounts to the coefficient vectors of each factor being orthogonal. Unitarity says this for the product. Section 4 transfers it to the factors with a degree argument: in a domain, a product can only be homogeneous if its factors are.

2 From the trajectory to a polynomial matrix

Exponential sums. Choose an orthonormal eigenbasis $e_1, \dots, e_{mn}$ of H with eigenvalues $\lambda_1, \dots, \lambda_{mn}$, repeats allowed, and put $v_j = \langle e_j, a \otimes b \rangle\, e_j$. Then $e^{-itH}(a \otimes b) = \sum_j e^{-i\lambda_j t} v_j$ with the v_j pairwise orthogonal. Identify $\mathbb{C}^m \otimes \mathbb{C}^n$ with $m \times n$ matrices, so that $a \otimes b$ becomes ab^{T} and the inner product becomes $\langle X, Y \rangle = \sum_{i,l} \overline{X_{il}} Y_{il}$. Write V_j for the matrix of v_j. The trajectory is

$$M(t) = \sum_j e^{-i\lambda_j t} V_j, \qquad \langle V_j, V_{j'} \rangle = 0 \ (j \neq j'), \tag{1}$$

and a vector is a product vector exactly when its matrix has rank at most one. Everything from here on uses only (1), and Sections 2 to 4 prove the following statement about it.

Theorem 4. *Let $M(t)$ be as in (1) and suppose $\operatorname{rank} M(t) \leq 1$ for every real t. Then there are finite families of vectors $x_\mu \in \mathbb{C}^m$ and $y_\nu \in \mathbb{C}^n$, each family pairwise orthogonal, and real numbers f_μ, g_ν, such that for all t*

$$M(t) = \Big(\sum_\mu e^{-if_\mu t} x_\mu \Big) \Big(\sum_\nu e^{-ig_\nu t} y_\nu \Big)^{\mathsf{T}}.$$

The orthogonality in the conclusion is the whole point. Without it, the factors of a rank-one matrix are just some curves in $\mathbb{C}^m$ and $\mathbb{C}^n$, and nothing says they are trajectories of a Hamiltonian.

The frequency lattice.

Lemma 5. *There are $k \geq 0$, real numbers $\omega_1, \dots, \omega_k$ linearly independent over $\mathbb{Z}$, vectors $e_j \in \mathbb{N}^k$ and $c \in \mathbb{R}$ with $\lambda_j = e_j \cdot \omega + c$ for every j.*

Proof. The additive group generated by the λ_j is a finitely generated torsion-free abelian group, hence free; let $\omega_1, \dots, \omega_k$ be a basis and $n_j \in \mathbb{Z}^k$ the coordinates of λ_j. With $N_i = \sum_j |n_{ji}|$ the vectors $e_j = n_j + N$ lie in $\mathbb{N}^k$, and $c = -N \cdot \omega$. □

Let $P = \mathbb{C}[X_1, \dots, X_k]$ and define the polynomial matrix

$$F = \sum_j X^{e_j} V_j \in P^{m \times n}, \qquad X^e = X_1^{e_1} \cdots X_k^{e_k}.$$

For $s \in \mathbb{N}^k$ let $N_s = \sum_{j:\, e_j = s} V_j$ be the coefficient of X^s in F. Distinct s collect disjoint sets of indices j, so the coefficient matrices N_s are pairwise orthogonal. On the torus point $z(t) = (e^{-i\omega_1 t}, \dots, e^{-i\omega_k t})$ we have $z(t)^{e_j} = e^{-i(e_j \cdot \omega)t}$, hence

$$M(t) = e^{-ict} F(z(t)). \tag{2}$$

In words, the trajectory is the polynomial matrix F evaluated along the line $t \mapsto z(t)$ on the torus, times a global phase.

Lemma 6. *If $p \in P$ and $p(z(t)) = 0$ for every real t, then $p = 0$.*

Proof. Write $p = \sum_r c_r X^r$. Then $p(z(t)) = \sum_r c_r e^{-i(r \cdot \omega)t}$. The map $r \mapsto r \cdot \omega$ is injective on $\mathbb{N}^k$ because the ω_i are independent over $\mathbb{Z}$, so the frequencies are distinct, and distinct characters $t \mapsto e^{-i\nu t}$ of $(\mathbb{R}, +)$ are linearly independent over $\mathbb{C}$ (Dedekind's lemma on independence of characters [4]). Hence every c_r vanishes. □

The 2×2 minors of $M(t)$ vanish for every t. By (2) each minor of $M(t)$ is e^{-2ict} times the corresponding minor of F evaluated at $z(t)$, so by Lemma 6 *every 2×2 minor of F is the zero polynomial*. We have traded a condition at every time for an identity in a polynomial ring.

3 Factoring in the polynomial ring

Lemma 7. *Let R be a GCD domain and $F \in R^{m\times n}$ a matrix whose 2×2 minors all vanish. Then $F = \alpha\beta^{\mathsf{T}}$ for some $\alpha \in R^m$ and $\beta \in R^n$.*

Proof. If $F = 0$ take $\alpha = \beta = 0$. Otherwise pick a column l_0 with an entry $F_{i_0 l_0} \neq 0$, let d be the gcd of that column and write $F_{il_0} = d\,\alpha_i$, so that $\gcd_i \alpha_i = 1$ and $\alpha_{i_0} \neq 0$. For any i, i', l the minor on rows i, i' and columns l, l_0 gives $F_{il}\, d\alpha_{i'} = d\alpha_i\, F_{i'l}$, so $F_{il}\alpha_{i'} = \alpha_i F_{i'l}$. Taking $i = i_0$, the element α_{i_0} divides $F_{i_0 l}\alpha_{i'}$ for every i', hence divides their gcd, which is $F_{i_0 l}\gcd_{i'}\alpha_{i'} = F_{i_0 l}$ up to a unit. Write $F_{i_0 l} = \alpha_{i_0}\beta_l$. Then $\alpha_{i_0}F_{il} = \alpha_i F_{i_0 l} = \alpha_{i_0}\alpha_i\beta_l$, and cancelling α_{i_0} gives $F_{il} = \alpha_i\beta_l$. □

A polynomial ring in finitely many variables over a field has unique factorisation (Gauss's lemma [4]) and is therefore a GCD domain. So $F = \alpha\beta^{\mathsf{T}}$ with $\alpha \in P^m$, $\beta \in P^n$ polynomial vectors. Write their coefficients as

$$\alpha = \sum_{\mu} X^{\mu} x_{\mu}, \qquad \beta = \sum_{\nu} X^{\nu} y_{\nu}, \qquad x_{\mu} \in \mathbb{C}^m,\ y_{\nu} \in \mathbb{C}^n.$$

It remains to show that the x_{μ} are pairwise orthogonal, and likewise the y_{ν}.

4 A grading argument

Doubling the variables. Take two sets of variables $u = (u_1, \dots, u_k)$ and $v = (v_1, \dots, v_k)$ and the ring $Q = \mathbb{C}[u, v]$. For a vector γ of polynomials in P let $\bar{\gamma}(u)$ be γ with conjugated coefficients, written in the variables u, and $\gamma(v)$ the same vector written in v. Define

$$G(\gamma) = \sum_i \bar{\gamma}_i(u)\,\gamma_i(v) \in Q.$$

If $\gamma = \sum_{\mu} X^{\mu} c_{\mu}$ with c_{μ} vectors, then

$$G(\gamma) = \sum_{\mu,\nu} \langle c_{\mu}, c_{\nu}\rangle\, u^{\mu} v^{\nu}. \tag{3}$$

In words: $G(\gamma)$ is the Gram matrix of the coefficient vectors of γ, spread out as a polynomial, with the entry $\langle c_{\mu}, c_{\nu}\rangle$ sitting on its own monomial $u^{\mu}v^{\nu}$. The coefficient vectors of γ are pairwise orthogonal exactly when $G(\gamma)$ has only the diagonal monomials $u^{\mu}v^{\mu}$.

Apply (3) to the entries of F, viewed as one long vector, and to α and β. Because $F_{il} = \alpha_i\beta_l$,

$$G(\alpha)\,G(\beta) = \sum_{i,l} \bar{F}_{il}(u) F_{il}(v) = G(F) = \sum_{s,s'} \langle N_s, N_{s'}\rangle u^s v^{s'} = \sum_s \|N_s\|^2 u^s v^s, \tag{4}$$

the last step by orthogonality of the N_s. This is the one place where unitarity of the evolution enters: the right side has only diagonal monomials.

Weights. For each coordinate $j \in \{1, \dots, k\}$ give u_j the weight -1, v_j the weight $+1$, and every other variable the weight 0. The monomial $u^{\mu}v^{\nu}$ then has weight $\nu_j - \mu_j$. The right side of (4) is homogeneous of weight 0 for every j.

Lemma 8. *Let $R = \mathbb{C}[Y_1, \dots, Y_r]$ be graded by integer weights $w(Y_i)$. If $p, q \in R$ are nonzero and pq is homogeneous, then p is homogeneous.*

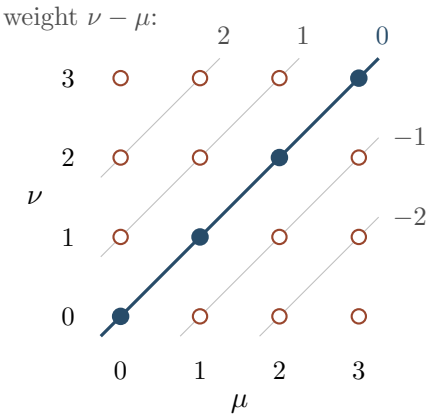


Figure 1: The monomials $u^\mu v^\nu$ of $G(\alpha)$ for $k = 1$, placed at (μ, ν). The weight $\nu - \mu$ is constant along each diagonal. A homogeneous $G(\alpha)$ lives on one diagonal; the nonzero diagonal entries $\|x_\mu\|^2$ (solid) put it on the main one, so every off-diagonal entry $\langle x_\mu, x_\nu \rangle$ (hollow) must vanish.

Proof. Let a and a' be the largest and smallest weights of monomials of p, and b, b' those of q. The weight-$(a+b)$ part of pq is the product of the weight-a part of p and the weight-b part of q, since any other pair of parts has smaller total weight. Both are nonzero and R is a domain, so pq has a monomial of weight $a + b$. In the same way it has one of weight $a' + b'$. As pq is homogeneous, $a + b = a' + b'$; with $a \geq a'$ and $b \geq b'$ this forces $a = a'$. □

Proposition 9. *Let $F \in P^{m \times n}$ have pairwise orthogonal coefficient matrices and vanishing 2×2 minors. Then $F = \alpha\beta^{\mathsf{T}}$ where the coefficient vectors of α are pairwise orthogonal, and so are those of β.*

Proof. Lemma 7 gives $F = \alpha\beta^{\mathsf{T}}$. If α or β is zero then $F = 0$ and there is nothing to prove, so assume both are nonzero. Some coefficient x_{μ_0} of α is nonzero, and by (3) the monomial $u^{\mu_0} v^{\mu_0}$ of $G(\alpha)$ has coefficient $\sum_i |x_{\mu_0, i}|^2 > 0$; in particular $G(\alpha) \neq 0$, and likewise $G(\beta) \neq 0$. Fix a coordinate j. By (4) and Lemma 8, $G(\alpha)$ is homogeneous for the j-th weight, and the monomial $u^{\mu_0} v^{\mu_0}$ shows that its weight is 0. So every monomial $u^\mu v^\nu$ with $\langle x_\mu, x_\nu \rangle \neq 0$ has $\nu_j = \mu_j$. This holds for every j, so $\langle x_\mu, x_\nu \rangle \neq 0$ forces $\mu = \nu$ (Figure 1). The same argument applied to $G(\beta)\, G(\alpha)$ handles β. □

5 Proof of the theorems

Proof of Theorem 4. Section 2 produced F with orthogonal coefficient matrices and vanishing minors. Proposition 9 gives $F = \alpha\beta^{\mathsf{T}}$ with orthogonal coefficient vectors x_μ and y_ν. By (2),

$$M(t) = e^{-ict} \alpha(z(t)) \beta(z(t))^{\mathsf{T}} = \Big(\sum_\mu e^{-i(\mu \cdot \omega + c)t} x_\mu \Big) \Big(\sum_\nu e^{-i(\nu \cdot \omega) t} y_\nu \Big)^{\mathsf{T}},$$

which is the claim with $f_\mu = \mu \cdot \omega + c$ and $g_\nu = \nu \cdot \omega$. □

Proof of Theorem 1. Apply Theorem 4 and drop the zero vectors from both families. Put

$$h_A = \sum_\mu f_\mu \frac{x_\mu x_\mu^*}{\|x_\mu\|^2}, \qquad h_B = \sum_\nu g_\nu \frac{y_\nu y_\nu^*}{\|y_\nu\|^2}.$$

These are Hermitian, and since the x_μ are pairwise orthogonal, $h_A x_\mu = f_\mu x_\mu$ and hence $e^{-ith_A} x_\mu = e^{-if_\mu t} x_\mu$; the same holds for h_B. With $\hat{a} = \sum_\mu x_\mu$ and $\hat{b} = \sum_\nu y_\nu$ this gives

$M(t) = e^{-ith_A}\hat{a}\,(e^{-ith_B}\hat{b})^{\mathsf{T}}$. At $t = 0$ we get $\hat{a}\hat{b}^{\mathsf{T}} = ab^{\mathsf{T}}$, so $\hat{a} = \gamma a$ and $\hat{b} = \gamma^{-1}b$ for some $\gamma \neq 0$, and the scalars cancel in the tensor product. □

Remark. The proof is constructive once the spectrum of H is known exactly. It also shows a little more: each factor of the trajectory is an exponential sum whose frequencies lie in the group generated by the eigenvalues of H, up to one common shift.

6 Constant entanglement is not enough

A product state has the smallest possible entanglement. It is natural to ask whether Theorem 1 survives when "stays a product state" is replaced by "keeps its entanglement", that is, keeps its Schmidt coefficients, so that $e^{-itH}\psi_0$ stays in the orbit of ψ_0 under local unitaries. For two qubits the answer depends on the amount of entanglement. At the other extreme from product states it is yes.

Theorem 10. *Let H be Hermitian on $\mathbb{C}^2 \otimes \mathbb{C}^2$ and let ψ_0 be maximally entangled. If $e^{-itH}\psi_0$ is maximally entangled for every $t \in \mathbb{R}$, then there are Hermitian h_A and h_B with $e^{-itH}\psi_0 = (e^{-ith_A} \otimes e^{-ith_B})\psi_0$ for all t.*

Proof. Write the trajectory as $M(t) = \sum_{s\in S} e^{-ist}W_s$ with distinct frequencies s and nonzero, pairwise orthogonal W_s (Section 2). Maximal entanglement means $M(t)M(t)^* = rI$; taking traces, $2r = \|M(t)\|^2 = \sum_s \|W_s\|^2 =: q$, a constant. Hence $|\det M(t)|^2 = \det(MM^*) = (q/2)^2$ is constant too. Now $\det M(t)$ is a finite exponential sum; if it had two or more frequencies with nonzero coefficients, the largest minus the smallest would appear in $|\det M(t)|^2$ with a nonzero coefficient, so $\det M(t) = \delta e^{-i\kappa t}$ for some $\delta \neq 0$ and real κ.

Choose w with $w^2 = (q/2)/\delta$, so $|w| = 1$, and put $N(t) = we^{i\kappa t/2}M(t) = \sum_s e^{-i(s-\kappa/2)t}W'_s$ with $W'_s = wW_s$. Then $NN^* = \frac{q}{2}I$ and $\det N = \frac{q}{2}$, so the adjugate of N equals $\det N \cdot N^{-1} = N^*$. Writing this out entrywise says $\tau(N(t)) = N(t)$ for the antiunitary involution

$$\tau\begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix} = \begin{pmatrix} \bar{x}_{11} & -\bar{x}_{10} \\ -\bar{x}_{01} & \bar{x}_{00} \end{pmatrix},$$

whose fixed points are the real multiples of $SU(2)$. Applying τ to the exponential sum conjugates the phases, so comparing coefficients of distinct characters gives a pairing of the frequencies: $\kappa - s \in S$ and $\tau(W'_{\kappa-s}) = W'_s$ for every $s \in S$.

Let T be the Hermitian operator on 2×2 matrices with eigenvectors W'_s and eigenvalues $s-\kappa/2$, and zero on their orthogonal complement. The pairing gives $\tau T = -T\tau$. The matrices $E_0 = I$, $E_1 = iZ$, $E_2 = iY$, $E_3 = iX$ are fixed by τ and orthogonal with $\|E_a\|^2 = 2$, and the two relations force $\langle E_a, TE_b\rangle = 2i\,\omega_{ab}$ with ω real and antisymmetric. A direct computation with the products of Pauli matrices shows that $TX = AX + XC$ for all X, where $A = \alpha\cdot\sigma$ and $C = \gamma\cdot\sigma$ with

$$\alpha = \tfrac{1}{2}\big(\omega_{03}+\omega_{12},\ \omega_{02}-\omega_{13},\ \omega_{01}+\omega_{23}\big), \qquad \gamma = \tfrac{1}{2}\big(\omega_{03}-\omega_{12},\ \omega_{02}+\omega_{13},\ \omega_{01}-\omega_{23}\big);$$

this is the splitting $\mathfrak{so}(4) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ made explicit. Hence $A'W_s + W_sC = sW_s$ with $A' = A + \kappa/2$. In eigenbases of A' and C this says that every nonzero entry of W_s sits at a pair of eigenvalues $a_k + c_l = s$, so $e^{-itA'}W_s e^{-itC} = e^{-ist}W_s$. Summing over s gives $M(t) = e^{-itA'}M(0)e^{-itC}$, which is the claim with $h_A = A'$ and $h_B = C^{\mathsf{T}}$. □

Between the two extremes the statement fails, and the counterexample is the second example of Section 1 again.

Example. Let $H = \frac{1}{2}(X \otimes X + Y \otimes Y + Z \otimes Z) - \frac{1}{2}(Z \otimes 1 + 1 \otimes Z) + \frac{3}{2}$ and $\psi_0 = -|00\rangle + (1 + i)|01\rangle + (i - 1)|10\rangle + |11\rangle$. The eigenvalues of H are $0, 1, 2, 3$, on the singlet, $|00\rangle$, the triplet state $(|01\rangle + |10\rangle)/\sqrt{2}$ and $|11\rangle$. With $z = e^{-it}$ the trajectory in matrix form is

$$M(t) = \begin{pmatrix} -z & 1 + iz^2 \\ iz^2 - 1 & z^3 \end{pmatrix}, \qquad \det M(t) = 1, \qquad \|M(t)\|^2 = 6.$$

So the squared Schmidt coefficients stay equal to $\frac{1}{2} \pm \frac{\sqrt{2}}{3}$ for all time and the concurrence stays $1/3$. Yet no time-independent local Hamiltonian generates this trajectory. If $M(t) = e^{-itA} M(0) e^{-itC}$, then $\det M(t) = e^{-it(\operatorname{tr} A + \operatorname{tr} C)} \det M(0)$, and constancy forces $\operatorname{tr} A + \operatorname{tr} C = 0$. The frequencies of such a trajectory lie among the sums $a_i + c_j$ of eigenvalues, a set that is symmetric under negation, since $(a_1 + c_1) + (a_2 + c_2) = (a_1 + c_2) + (a_2 + c_1) = 0$. The trajectory above has frequencies $0, 1, 2, 3$, each with a nonzero component, and no set of at most four reals symmetric under negation contains them.

The mechanism is the resonance $1 + 3 = 2 + 2$. Expanding, $\det M(t) = (-z)(z^3) - (1 + iz^2)(iz^2 - 1) = -z^4 - (-1 - z^4)$: the term from the $|00\rangle$ and $|11\rangle$ components, at frequency $1 + 3$, cancels the square of the triplet component, at frequency $2 + 2$. This is exactly the kind of cancellation that Theorem 1 had to rule out for product states. There, rank one leaves no room for it; at intermediate entanglement there is room.

7 Related questions

The question [8] was posed as a simplification of two harder ones. The first replaces product states by entangled states whose entanglement does not change in time; Section 6 shows that the answer is then no in general, already for two qubits, and yes for two qubits that stay maximally entangled. The second asks whether a Hamiltonian that acts "simply" on every state of a subspace acts simply on the subspace as a whole [7], and remains open as far as we know. A related question about non-entangling bases of bipartite unitaries is [6].

8 Formal verification

The Lean 4 development [5] proves Theorem 4 in the form stated above, from the hypotheses (1) (an arbitrary finite family of reals λ_j, repeats allowed, and pairwise orthogonal matrices V_j) and the hypothesis that $M(t)$ is a column times a row for every real t. It contains Lemmas 5 to 8 and Proposition 9, with Dedekind's lemma, unique factorisation of polynomial rings and the existence of bases of finitely generated free abelian groups taken from Mathlib. The main theorem depends only on the standard axioms (propositional extensionality, choice and quotient soundness). Mutants, each weakening one hypothesis or strengthening one conclusion, fail to check (counted below). The two steps outside the formal development are standard linear algebra: the spectral decomposition at the start of Section 2, and the construction of h_A and h_B from orthogonal vectors in the proof of Theorem 1, together with the induction of Corollary 2. A short computation in exact rational arithmetic, independent of the Lean code, runs the construction on degenerate and resonant examples with one and two independent frequencies.

Section 6 is formalised as well. Theorem 10 is proved in the exponential-sum form used in its proof, with the same route: the single-frequency lemma for constant-modulus sums, the involution τ, the pairing by comparison of coefficients, the explicit splitting of $\mathfrak{so}(4)$, and the evolution of eigenvectors of $X \mapsto AX + XC$. For the Example, Lean checks the eigen-equations of H in Pauli form, the expansion of $M(t)$ into orthogonal components, $\det M(t) = 1$ and $\|M(t)\|^2 = 6$, and that no Hermitian A and C satisfy $M(t) = e^{-itA} M(0) e^{-itC}$ for all t, using

the matrix exponential of Mathlib. All of these depend only on the standard axioms, and twenty-six mutants in total fail to check. The Lean files and both exact computations are at [1].

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